WEEK 02 | CYBERSECURITY | NETWORKWALKS

PENETRATION TESTING REPORT

FOOTPRINTING, NETWORK SCANNING & VULNERABILITY ASSESSMENT

Engagement Parameter	Details
Pentester Name (Cybersecurity Professional)	Maloni Zulu
Program/Batch	B084-Networkwalks
Date	12 November 2026
Modules completed	Week 02: Footprinting (Multiple Kali Tools) Week 02: Nmap & Zenmap Scanning
Client/Target	networkwalks.com (written permission secured)
Permission secured from client?	Yes
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phase 3-5: In Progress

1. Liability Disclaimer

> This project was conducted strictly for educational and cybersecurity training purposes. I performed these activities only on systems where I had secured written permission or that are explicitly provided for security testing. No exploitation or unauthorized access was attempted.
>
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2. Introduction

This report documents the practical activities carried out during Week 2 of my Cybersecurity Internship at Networkwalks. It covers footprinting the `networkwalks.com` domain with multiple Kali Linux tools, and scanning it with Nmap and Zenmap. Together they show how an attacker moves from gathering public information to mapping live services on a network.

All commands were run in Kali Linux (footprinting and Nmap), installed in Oracle VirtualBox 7.2 on a Windows 11 Pro host (Intel Core i5, 8 GB RAM allocated), and Zenmap on Windows 11. Every step below includes the exact command used, what I observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view. The goal was not only to run the tools but to interpret their output and judge the security implications.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux & Windows 11	Operating systems used (Kali in Oracle VirtualBox on a Windows 11 host).
WHOIS	Find domain registration details (registrar, dates, name servers).
WhatWeb	Fingerprint web technologies (server, CMS, plugins).
Nslookup	Resolve the domain name to its IP address using DNS.
cURL (-I)	Read the HTTP response headers of the website.
WAFW00F	Detect whether a Web Application Firewall protects the site.
DNSRecon	Enumerate DNS records (NS, MX, SPF, TXT).
Nmap	Discover open ports, service versions and OS fingerprints.
Zenmap (Nmap GUI)	Run an Intense Scan and view ports, hosts and network topology.

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance against `networkwalks.com` using six Kali Linux tools: WHOIS, WhatWeb, Nslookup, cURL, WAFW00F and DNSRecon. Each was used to collect a different type of information.

I first used WHOIS (`whois networkwalks.com`) to retrieve registration details. The registrar is GoDaddy, the domain was created on 6 Nov 2019 and expires on 6 Nov 2027, DNSSEC is unsigned, and the name servers are `NS6135.HOSTGATOR.COM` and `NS6136.HOSTGATOR.COM`, which identifies HostGator hosting. Transfer, update and delete lock statuses are set.

I then used WhatWeb (`whatweb networkwalks.com` and `whatweb -v`) to fingerprint the site. It identified WordPress 7.1.4 and the WP Download Manager 3.3.76 plugin, plus Apache, jQuery 3.7.1, Google Tag Manager, the server IP `192.232.216.135` and the contact email `info@networkwalks.com`.

Using Nslookup (`nslookup networkwalks.com`), I resolved the domain to `192.232.216.135`. My first DNS server (`10.120.145.111`) timed out, but a non-authoritative answer was still returned. A direct IP suggests the site is not hidden behind a CDN.

I used cURL with the `-I` option (`curl -I https://networkwalks.com`) to inspect response headers. The server returned HTTP 200 OK (HTTP/2) and showed `server: Apache`, a WordPress cache header and the WordPress REST API endpoint `/wp-json/`. The response did not show Strict-Transport-Security or Content-Security-Policy headers.

Next, WAFW00F (`wafw00f networkwalks.com`) identified ModSecurity (SpiderLabs) as the Web Application Firewall protecting the site (WAFW00F v2.3.2, 2 requests).

Finally, DNSRecon (`dnsrecon -d networkwalks.com`) enumerated NS, MX, TXT and SPF records. It also reported BIND 9.16.23 and cPanel-related records.

> [Check before submitting: the DNSRecon screenshot (Figure 6) shows the domain mistyped as networlwalks.com and the run failed with 'Could not resolve domain'. Re-run dnsrecon -d networkwalks.com, replace Figure 6 and confirm the findings above.]

4.2 Network Scanning with Nmap & Zenmap

For the second activity I used Nmap on Kali (`nmap -T4 -A -v networkwalks.com`) and then Zenmap with the Intense Scan profile, which runs the same command and adds service detection, OS detection and traceroute. The host (`192.232.216.135`) was up, and Nmap scanned 1000 ports and found 15 open ports and 985 closed ports.

The services found are listed below.

Port(s)	Service	Version
21	ftp	Pure-FTPd (with SSL/TLS)
22 / 2222	ssh	OpenSSH 9.9 (protocol 2.0)
25 / 26	smtp	Exim smtpd 4.100.1
53	domain	Generic DNS response (NOTIMP)
80 / 443	http / https	Apache httpd
110 / 995	pop3 / pop3s	Dovecot pop3d
143 / 993	imap / imaps	Dovecot imapd
465	smtps	Not identified
587	submission	tcpwrapped
3306	mysql	MySQL 5.7.44-48

Zenmap's other views: Ports / Hosts listed the services above in one table. Topology showed the route from my machine (localhost) through my local gateway `10.120.145.111` and several ISP and backbone hops (including ldn and ewr Twelve99 routers) to the target. Host Details reported the OS as Linux 2.6.28 and a PTR hostname on `unifiedlayer.com`. Nmap's OS guess is an estimate, so it should not be treated as exact.

4.3 Errors Encountered and Solutions

Error / Challenge	Possible Cause	Solution
DNSRecon command not found	Tool was not installed	Installed with `sudo apt install dnsrecon`
WHOIS timeout	Network or server response delay	Checked connectivity and retried
cURL connection delay	Slow network or HTTPS handshake delay	Checked connectivity and retried
WAFW00F / Nmap / Zenmap scan delay	Many requests, filtering, Intense Scan runs several detections	Reviewed scan options and let the scan finish

Lesson learned: understanding why a tool fails makes it easier to fix and to trust the results.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Public WHOIS information	WHOIS showed registrar, dates and HostGator name servers; DNSSEC unsigned	Can support infrastructure reconnaissance	• Low
2	Web technology information exposed	WhatWeb identified WordPress 7.1.4 and WP Download Manager 3.3.76	Exposed versions can be cross-referenced against known advisories	• Medium
3	Server IP address identifiable	Nslookup and WhatWeb showed 192.232.216.135	Provides information about hosting infrastructure; no CDN in front	• Low
4	HTTP technical information exposed	cURL showed server: Apache, /wp-json/ and no HSTS or CSP header	May reveal configuration details and missing security controls	• Medium
5	WAF technology identifiable	WAFW00F identified ModSecurity (SpiderLabs)	Reveals information about the defensive architecture	• Low
6	DNS infrastructure information exposed	DNSRecon records (to be re-confirmed, see Figure 6)	Can assist infrastructure enumeration	• Medium
7	Database port exposed to the internet	Nmap found 3306/tcp open running MySQL 5.7.44-48	A public database port widens the attack surface; MySQL 5.7 is end-of-life and no longer receives security fixes	• High
8	Many network services exposed	15 open ports: FTP, two SSH ports, SMTP, DNS, POP3, IMAP and more	Each exposed service is an extra target; unused ones should be closed or restricted	• Medium

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
9	Operating system fingerprinting	Zenmap reported Linux 2.6.28 (an estimate)	May help identify platform-specific attack opportunities	● Low

Risk level key: ● High ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities. No exploitation or vulnerability validation was performed. The presence of a software version, IP address, open port or DNS record does not by itself mean a system is vulnerable. The priorities are preliminary and need further authorized testing to confirm.

6. Recommendations

Based on these observations, I recommend the following security improvements:

- **Review public domain information:** Review what registration data is publicly visible and enable domain privacy where appropriate.
- **Keep software updated:** Keep WordPress, plugins and other web components updated and checked against current security advisories.
- **Reduce technology exposure:** Review what CMS, plugin and version information is visible to the public.
- **Review HTTP headers:** Remove unnecessary server information and add security headers such as Content-Security-Policy, Strict-Transport-Security and X-Content-Type-Options.
- **Protect the origin server:** Apply network controls so the origin server is protected, for example by restricting direct access.
- **Maintain the WAF:** Keep ModSecurity rulesets updated and review its security logs regularly.
- **Audit DNS records:** Check DNS records periodically and remove obsolete entries so only required services are exposed.
- **Restrict the database port:** Do not expose MySQL (3306) to the internet; limit it to localhost or trusted IPs, and upgrade from the end-of-life MySQL 5.7.
- **Manage ports and services:** Disable unnecessary services (such as FTP if unused, and the second SSH port) and restrict access to the rest.
- **Harden operating systems:** Maintain current security patches and apply host firewall policies.
- **Test only with authorization:** Reconnaissance and scanning should only be performed where appropriate authorization has been given.

7. Conclusion

Week 2 of my Cybersecurity Internship at Networkwalks gave me practical experience in footprinting, reconnaissance, network scanning and security analysis. I worked with WHOIS, WhatWeb, Nslookup, cURL, WAFW00F, DNSRecon and Nmap, and used Zenmap to view the scan results graphically.

I learned that reconnaissance is a key step in understanding an organization's digital footprint before deeper testing. WHOIS and DNSRecon give domain and DNS intelligence, WhatWeb reveals technologies, cURL exposes headers, WAFW00F shows defensive filtering, and Nmap and Zenmap reveal ports, services and network paths.

I also learned that an open port, exposed technology or public record does not automatically mean a vulnerability exists. Findings need validation and context before a risk is assigned, and they must be documented clearly: what was done, what was found, what it means and how to reduce the risk. Finally, all scanning must stay within an authorized scope, as it did here.

8. Evidences Collected



Figure 1: WHOIS lookup (whois networkwalks.com)

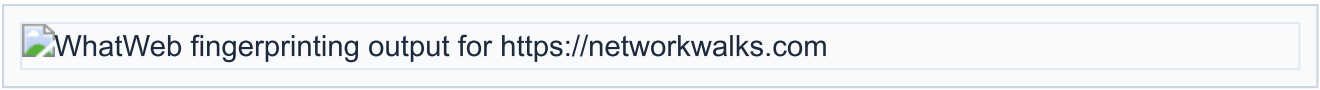

WhatWeb fingerprinting output for https://networkwalks.com

Figure 2: WhatWeb fingerprinting (whatweb https://networkwalks.com)


Nslookup resolution output for networkwalks.com

Figure 3: Nslookup resolution (nslookup networkwalks.com)


cURL headers output for https://networkwalks.com

Figure 4: cURL headers (curl -I https://networkwalks.com)


WAFW00F detection output for networkwalks.com

Figure 5: WAFW00F detection (wafw00f networkwalks.com)


DNSRecon command screenshot

Figure 6: DNSRecon (dnsrecon -d ...) - re-run with the correct domain


Nmap scan output for networkwalks.com

Figure 7: Nmap scan (nmap -T4 -A -v networkwalks.com)


Zenmap Intense Scan output

Figure 8: Zenmap Intense Scan output


Zenmap Ports / Hosts view

Figure 9: Zenmap Ports / Hosts view


Zenmap Topology view

Figure 10: Zenmap Topology view


Zenmap Host Details OS match

Figure 11: Zenmap Host Details (OS match)